

Find the expected value of the random variable.

x	2	3	4	5
P(x)	0.2	0.3	0.1	0.4

If the random variable x can take on the n values $x_1, x_2, x_3, \dots, x_n$ with probabilities $p_1, p_2, p_3, \dots, p_n$ respectively, then the expected value is $E(x) = x_1 p_1 + x_2 p_2 + x_3 p_3 + \dots + x_n p_n$.

Here the random variable x can take on the values 2, 3, 4, and 5.

$$\begin{aligned} E(x) &= 2(0.2) + 3(0.3) + 4(0.1) + 5(0.4) \\ &= 0.4 + 0.9 + 0.4 + 2.0 \\ &= 3.7 \end{aligned}$$